

EXERCISE I

In reference to the Strassen Matrix Multiplication technique

I.a) Show the calculation of the product of the terms P1-P7 for the matrix product $[[1,2][3,4]] \times [[5,6][7,8]]$

$$M_1 = (A_{11} + A_{22})(B_{11} + B_{22});$$

$$M_2 = (A_{21} + A_{22})B_{11};$$

$$M_3 = A_{11}(B_{12} - B_{22});$$

$$M_4 = A_{22}(B_{21} - B_{11});$$

$$M_5 = (A_{11} + A_{12})B_{22};$$

$$M_6 = (A_{21} - A_{11})(B_{11} + B_{12});$$

$$M_7 = (A_{12} - A_{22})(B_{21} + B_{22}),$$

using only 7 multiplications (one for each M_k) instead of 8. We may now express the C_{ij} in terms of M_k :

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} M_1 + M_4 - M_5 + M_7 & M_3 + M_5 \\ M_2 + M_4 & M_1 - M_2 + M_3 + M_6 \end{bmatrix}.$$

Where M# corresponds to P# in our case

$$P1 = (a_{11} + a_{22})(b_{11} + b_{22}) = (1+4)*(5+8)=65$$

$$P2 = (a_{21} + a_{22})b_{11} = (3+4)*5=35$$

$$P3 = a_{11}(b_{12} - b_{22}) = 1*(6-8)=-2$$

$$P4 = a_{22}(b_{21} - b_{11}) = 4*(7-5)=8$$

$$P5 = (a_{11} + a_{12})b_{22} = (1+2)*8=24$$

$$P6 = (a_{21} - a_{11})(b_{11} + b_{12}) = (3-1)*(5+6)=22$$

$$P7 = (a_{12} - a_{22})(b_{21} + b_{22}) = (2-4)*(7+8)=-30$$

$$C_{11} = P1 + P4 - P5 + P7 = 19$$

$$C_{12} = P3 + P5 = 22$$

$$C_{21} = P2 + P4 = 43$$

$$C_{22} = P1 - P2 + P3 + P6 = 50$$

I.b) estimate the number multiplications and additions needed to compute the product of two 4x4 matrices with the Strassen method (use the recursive formula)

We can consider a 4x4 matrix as composed of 4 2x2 matrices so considering the recursive formula we get

MUL(N), ADD(N) – multiplication/addition of matrices of size NxN

$$\text{MUL}(N) = 7 * \text{MUL}(N / 2) + 18 * \text{ADD}(N/2)$$

$$\text{MUL}(4) = 7 * \text{MUL}(2) + 18 * \text{ADD}(2)$$

$$\text{MUL}(2) = 7 * \text{MUL}(1) + 18 * \text{ADD}(1) = 7\text{MUL} + 18\text{ADD}$$

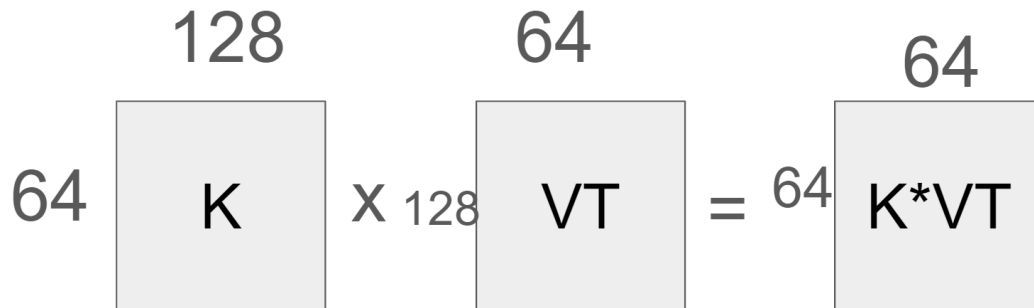
$$\text{Tot_strassen: } 7 * (7\text{MUL} + 18\text{ADD}) + 18 * 4\text{ADD} = 49\text{MUL} + 198\text{ADD}$$

I.c) Using the asymptotic approximation compute the number of operations for the same matrix product of (I.b). Explain why there are differences, if any.

The asymptotic formula is $N^{2.81}$, so it leads to $4^{2.81} \approx 49.18$, approximating to the nearest integer gives 49, as operations cannot be fractional, which is the correct amount.

EXERCISE II

Consider a simplified attention module $A = \text{SoftMax}(K \cdot VT)$, where K is 64×128 matrix and VT is 128×64 . Matrix elements in FP16 format



II.a) Computed the total memory space needed to store, K, VT, A

$$K = 64 * 128 * 2B = 16.384B$$

$$VT = 128 * 64 * 2B = 16.384B$$

$$A = 64 * 64 * 2B = 8.192B$$

$$\text{Total} = K + VT + A = \mathbf{40960 B}$$

II.b) Compute the total number of load and store operations (read and writes) needed for computing the attention module, assuming that matrix multiplication is computed with the trivial algorithms (triple nested loop iteration)

MatMul:

$$\text{w/ psum in memory: } 64 * 64 * 128 * 3 \text{ LOAD} + 64 * 64 * 128 \text{ STORE} = 1.572.864 \text{ LOAD} + 524.288 \text{ STORE}$$

$$\text{w/ psum in local reg.: } 64 * 64 * 128 * 2 \text{ LOAD} + 64 * 64 \text{ STORE} = 1.048.576 \text{ LOAD} + 4096 \text{ STORE}$$

Softmax:

Per row:

Denominator: 64 LOAD

Normalization: 64 LOAD + 64 STORE

Total Softmax: $64 * (\text{denominator} + \text{normalization}) = 64 * (64 \text{ LOAD} + 64 \text{ LOAD} + 64 \text{ STORE}) = 8192 \text{ LOAD} + 4096 \text{ STORE}$

Total Attention: MatMul + Softmax

Ver. A) MatMul w/ psum in memory + Softmax = **1.581.056 LOAD + 528.384 STORE**

Ver. B) MatMul w/ psum in local reg + Softmax = **1.056.768 LOAD + 8.192 STORE**

II.c) Compute the total number of elementary arithmetic operations needed to compute A, assuming that FMA operations count as 2 and complex (exponentiation, division) operations count as 10.

MatMul: $64 * 64 * 128 * 1 \text{ FMA} = 1.048.576 \text{ ops}$

Softmax w/o storing exp in mem:

Denominator: $64 * (64 \text{ EXP} + 63 \text{ ADD})$

Normalization: $64 * 64 * (1 \text{ EXP} + 1 \text{ DIV})$

Total Softmax: $8.192 \text{ EXP} + 4.032 \text{ ADD} + 4.096 \text{ DIV} = 126.912 \text{ ops}$

Softmax w/ storing exp in mem:

Denominator: $64 * (64 \text{ EXP} + 63 \text{ ADD})$

Normalization: $64 * 64 * 1 \text{ DIV}$

Total Softmax: $4.096 \text{ EXP} + 4.032 \text{ ADD} + 4.096 \text{ DIV} = 85.952 \text{ ops}$

Total: MatMul + Softmax

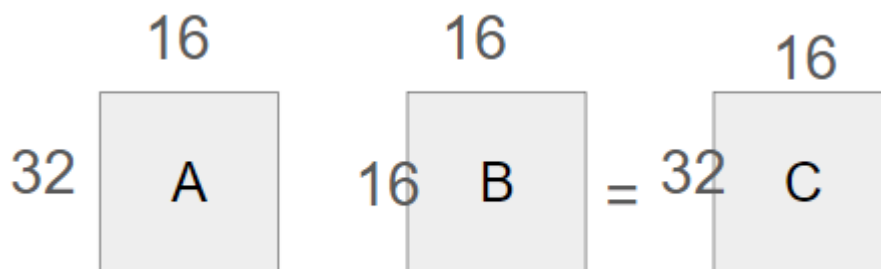
Ver. A) MatMul + Softmax w/o storing exp in mem: **1.175.488 ops**

Ver. B) MatMul + Softmax w/ storing exp in mem: **1.134.528 ops**

EXERCISE III

A parallel cluster of RISC-V processors is executing matrix multiplication followed by softmax of matrices A (32x16) and B (16x16). Assume that the matrix multiplication computation is parallelized assigning rows of the output matrix C to the processors. Following the matrix multiplication, the softmax is parallelized again assigning rows of the output matrix to processors. Assume a barrier between the two kernels with negligible overhead. The cluster has 12 processors. Elements of A, B, C are 16bit FP numbers

III.a) Compute the L1 minimal shared memory size that is needed to store the complete matrices, and the total number of MACs needed to complete the matrix multiplication.



L1 memory requirements is is equal to

$$A = 32 \times 16 \times 2 \text{ B} = 1024 \text{ B}$$

$$B = 16 \times 16 \times 2 \text{ B} = 512 \text{ B}$$

$$C = 16 \times 32 \times 2 \text{ B} = 1024 \text{ B}$$

$$\text{total memory is then } 1024 \text{ (A)} + 512 \text{ (B)} + 1024 \text{ (C)} = 2560 \text{ B}$$

$$\text{The number of MAC is } 16 \times 16 \times 32 = 8192$$

III.b) Compute the number of rows to be assigned to each core for the two kernels, trying to minimize the workload unbalance.

For each of the Kernels we have $32/12 = 2 + 8/12$ rows which means that 8 cores would get 3 rows and 4 get 2 rows

III.c) Compute the speedup with respect to sequential computation. Assume 10% of stall cycles due to banking conflicts in the parallel execution.

$$\text{Speed up} = \frac{n_{\text{rows}} \times n_{\text{kernels}}}{[n_{\text{kernel}} \times \max_{\text{number_rows_core}} \times (1 + \text{stalls})]} =$$

$$= \frac{32 \times 2}{[2 \times 3 \times (1 + 0.1)]} = 9.7 \times$$